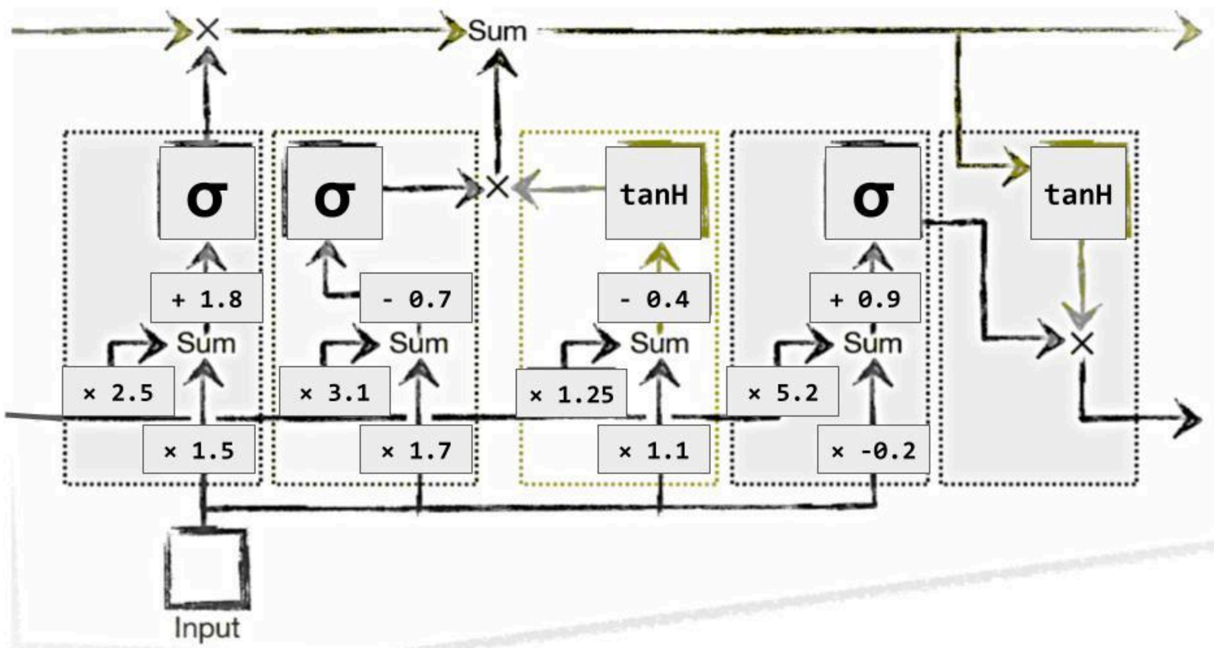
CSE427 Section 5 Quiz 4
Fall 2025
Duration: 25 Minutes

Name: **ID:**

Question *[10 Points]*

Consider the following LSTM unit.



Assume both the initial values of Long Term and Short Term memory values are set as 2. Now using two LSTM cells, predict the output value of Day 3 if input Day 1 and Day 2 values are 2 and 1.5 respectively.

Solution

Cell 1 (Day 1):	**Cell 2 (Day 2):**
Forget Gate:	**Forget Gate:**
$f1 = \sigma(9.8) \approx 0.9999446$	$f2 = \sigma(6.5375758) \approx 0.9985541$
Input Gate:	**Input Gate:**
$i1 = \sigma(8.9) \approx 0.9998636$	$i2 = \sigma(4.9345939) \approx 0.9928580$
$c1\sim = \tanh(4.3) \approx 0.9996319$	$c2\sim = \tanh(2.4937879) \approx 0.9864481$
$c1 \approx 2.9993846$ (Long Term)	**$c2 \approx 3.9744507$ (Long Term)**
Output Gate:	**Output Gate:**
$o1 = \sigma(10.9) \approx 0.9999815$	$o2 = \sigma(5.7741576) \approx 0.9969028$
$h1 \approx 0.9950303$ (Short Term)	**$h2 \approx 0.9961991$ (Short Term)**

Predicted output of Day 3 = **0.9962**